

Detailed Solutions - Unit 1 Probability

A. Probability - Conceptual

Sample space: set of all outcomes. Example: two dice $\rightarrow$ 36 outcomes.
Event: subset of sample space. Example: sum = 7.
Mutually exclusive vs independent events explained with dice/coins.
Conditional probability formula explained with example.
Bayes theorem explained as updating beliefs.

B. Probability - Applications

Coin tossed twice, probability at least one head = 0.75.
Two dice sum = 7 probability = $1/6$.
Card face given red probability = $3/13$.
Two-child family given one boy, probability both boys = $1/3$.
Box 3 red, 2 blue, 1 green, probability of drawing 2 red = $1/5$.

D. Probability - Problem Solving

One die shows 4, probability sum = 8 is $1/11$.
Urn with 4 white and 6 black, probability both black = $1/3$.
Hat problem with 3 players, probability none gets own hat = $1/3$.
Blood test example: probability disease given positive = 33.11 percent.

H. Probability - Calculations

Lottery 6/49, probability exactly 3 matches = 0.01765.
Factory Bayes example: 60 percent chance defective from Machine A.
Three children given elder boy, probability all boys = 0.25.
Poker full house probability = 0.00144.
Birthday problem with 25 people = 56.87 percent chance of shared birthday.

K. Random Variables - Applications

Coin until head follows geometric distribution.
Sum of two dice = 7 probability = $1/6$.
Geometric distribution $p=0.2$, probability $X=5$ = 0.08192.
Poisson distribution with mean 4, probability $X=2$ = 0.1465.
Normal(0, variance 4), probability between -1 and 1 = 0.383.

L. Random Variables - Problem Solving

Binomial(4,0.5), probability $X=2$ = 0.375.
Uniform(0,10), probability between 2 and 8 = 0.6.
Exponential $\lambda=0.5$, probability $X>3$ = 0.2231.
Geometric $p=0.3$, expectation = 3.33, variance = 7.78.
Joint PMF example: probability $X=1$ = 0.5, probability $X+Y=3$ = 0.3.